

PEI at #SMM4H-HeaRD 2025: Enhancing Adverse Event Detection via Error-Driven Data Augmentation

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Abstract

This paper outlines our contributions to Task 1 and Task 6 of the #SMM4H-HeaRD 2025 Workshop, both focused on the binary classification of social media text reporting adverse medical events. Task 1 involves detecting adverse drug events (ADEs) in multilingual texts written in German, French, Russian, and English. Task 6 targets the identification of adverse reactions to herpes zoster vaccines in Reddit posts. We employed transfer learning using the encoder-based RoBERTa model as our baseline, alongside the large language model (LLM) Mistral. Our focus was on enhancing LLM performance by introducing an iterative error-driven data augmentation strategy, in which false predictions were paraphrased using the LLM, added back to the training set, and retraining the model from scratch. This refinement loop was repeated twice for both tasks. Our augmentation strategy enabled the Mistral model to outperform the RoBERTa baseline, with the best-performing Mistral models achieving an F1 score of 0.96 in Task 6 and a macro-averaged F1 score of 0.709 across languages in Task 1, resulting in the highest performance among all participants in the Task 1 competition. To support third-party use, we have released our best-performing models for both Task 1 and Task 6 on Hugging Face.

1 Introduction

The #SMM4H-HeaRD 2025 Workshop, as outlined in the overview by Klein et al. (2025), challenged participants to detect ADEs and vaccine adverse event mentions (VAEM) in Task 1 and Task 6, respectively. Both tasks are framed as binary classification problems, but differ in terms of language diversity and platform characteristics. Task 1 focuses on ADE classification in multilingual social media content, including tweets, reviews, sentences, and posts, while Task 6 involves VAEM detection in English Reddit posts.

Previous #SMM4H-HeaRD workshops have also partially addressed ADE detection from social media, often focusing on pipelines combining classification, entity extraction, and normalization (Klein et al. 2023; Xu et al. 2024). As some of the shared datasets suffer from class imbalance, several recent studies have explored data augmentation

strategies to address this challenge. For instance, Ta et al. (2024) used ChatGPT to paraphrase the validation set and add it to the training data; however, this approach did not yield the best-performing model in their experiments. Similarly, Li et al. (2024) experimented with four different LLM-based augmentation techniques (using GLM-4 and GPT-3.5) for both ADE extraction and normalization.

Although some studies have shown that LLMs do not consistently outperform encoder-based models in classification tasks (Wang et al. 2023; Sun et al. 2023), there were efforts in the #SMM4H-HeaRD 2024 competition to enhance LLM performance beyond that of encoder-based models. For example, Khademi et al. (2024), working on the same dataset for Task 6, used Twitter-RoBERTa as a baseline and attempted to enhance GPT-3 and GPT-4 using prompt engineering and few-shot learning. Despite their efforts, the LLMs did not outperform the encoder-based model.

In our participation in the #SMM4H-HeaRD 2025 Workshop, we leveraged the encoder-based model twitter-*xlm-roberta-large-2022* (Barbieri, Anke, and Camacho-Collados 2022) as a baseline, and the LLM-based generative model Mistral (Mistral-Nemo-Instruct-2407)¹. To improve Mistral's performance, we applied an iterative error-driven data augmentation strategy, in which false predictions on the training data were paraphrased, added back to the training set, and the model was retrained from scratch. This method enhanced Mistral's performance across both tasks and allowed it to outperform the RoBERTa baseline.

2 Tasks and Datasets

2.1 Task 1

Task 1, titled “Binary Classification of Social Media Posts Containing Adverse Drug Events”, focuses on the identification of ADEs in multilingual social media texts, including tweets, reviews, sentences, and posts written in German,

¹ <https://huggingface.co/mistralai/Mistral-Nemo-Instruct-2407>
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French, Russian, and English. Given a text, the system must determine whether it contains a mention of an ADE (label "1") or not (label "0").

The training dataset for Task 1 consists of 31,187 multilingual social media posts, with an additional 4,625 samples in the validation set. The data spans four languages: German, French, Russian, and English. The German and French data were collected from lifeline.de, with the French set translated from German (Raithel et al. 2024; Raithel et al. 2022). The Russian data is a combination of the RuDReC drug review corpus (Tutubalina et al. 2020) and data from SMM4H 2021 Task 2 (Magge et al. 2021), while the English data was sourced from Twitter and provided in SMM4H 2024 (Xu et al. 2024). The largest subsets are in English (17,974 training samples) and Russian (10,754), followed by German (1,482) and French (977). The dataset is highly imbalanced, with ADE-positive examples comprising only around 1% of the total data. The test set includes 23,214 examples across all four languages; however, the labels are not publicly available and were used solely for official evaluation.

2.2 Task 6

Task 6, titled "Detection of Vaccine Adverse Event Mentions in Reddit Posts", targets the automatic classification of Reddit posts related to herpes zoster (shingles) vaccines. The goal is to distinguish between posts that include personal mentions of adverse vaccine reactions (positive class, "1") and those that discuss vaccines without reporting an adverse event (negative class, "0").

The dataset consists of 2,521 training samples and 786 validation samples, which were collected using the Python Reddit API Wrapper (PRAW) (Khademi et al. 2024). The distribution is relatively balanced in both sets, with 1,149 positive and 1,372 negative samples in the training set, and 366 positive and 420 negative samples in the validation set. The test set contains 8,106 posts, but the labels are not publicly available and were used exclusively for official evaluation on the organizers' platform.

3 Methodology

To select the baseline encoder model, we considered that the annotated data for Task 1 and Task 6 consists of social media content—multilingual text for Task 1 and English text for Task 6. Based on this, we used the RoBERTa-based model "twitter-xlm-roberta-large-2022". We also selected "Mistral-Nemo-Instruct-2407" as the LLM-based model, used for both binary classification and paraphrasing.

To enhance model performance, we developed an iterative error-driven data augmentation approach, inspired by

concepts from AdaBoost (Adaptive Boosting) (Freund and Schapire 1996). AdaBoost is a method designed to improve classification performance by focusing on misclassified training instances (false predictions). It sequentially trains multiple models, where each new model pays more attention to the errors made by the previous one. The final prediction is obtained through a weighted combination of all models. In contrast, our approach does not involve model ensembling or sample reweighting. Instead, we identify false predictions in the training data—often hard-to-learn examples—and paraphrase them to create semantically equivalent but lexically diverse variants. These augmented samples are added back to the training set, and the model is retrained from scratch on the updated dataset. This allows the model to learn from its earlier mistakes and better generalize to hard-to-learn examples within a single-model framework. This process was repeated for two iterations for both Task 1 and Task 6.

Regarding model evaluation metrics, as specified by the organizers, models were assessed using F1 score, precision, and recall for Task 6, and per-language F1 scores as well as the macro-averaged F1 score across languages for Task 1.

4 Experiments

We fine-tuned the RoBERTa-based model using a batch size of 32, for 3 epochs, and a maximum sequence length of 512. To address the significant class imbalance in Task 1—both across languages and between positive and negative samples—we set the learning rate to $1e-5$. For Task 6, a learning rate of $2e-5$ was used. All other parameters followed the default configuration from the original RoBERTa paper (Liu et al. 2019).

To fine-tune the Mistral model, we used four NVIDIA A100 GPUs (GA100, 20b5 rev A1, 80GB HBM2e), with a batch size of 1 per GPU. Fine-tuning was carried out using the official Mistral fine-tuning code, with a learning rate of $6e-5$, a sequence length of 4096, and all remaining parameters set to the defaults recommended on the official GitHub repository². During experimentation, we also designed and selected optimized prompts for binary classification in Task 1 and Task 6, which are provided in Appendix A and Appendix B, respectively.

For each fine-tuned Mistral model, we applied our iterative error-driven data augmentation strategy. The initial model was first evaluated on the training data to identify false predictions, which were then paraphrased using the Mistral model. These paraphrased examples were added back to the training set, and the model was retrained from

² <https://github.com/mistralai/mistral-finetune>

Model	Dataset	Dataset Size	On validation data				
			F1-en	F1-de	F1-fr	F1-ru	F1
Roberta	Training	31,187	0.729	0.676	0.781	0.717	0.726
Mistral	Training	31,187	0.835	0.557	0.815	0.704	0.728
Mistral	Training + Paraphrased FP (Iteration 1)	32,031	0.848	0.548	0.849	0.712	0.739
Mistral	Training + Paraphrased FP (Iterations 1 + 2)	32,894	0.817	0.667	0.844	0.717	0.761

Abbreviations: en = English, de = German (Deutsch), F1 = F1 Score, FP = False Predictions, ru = Russian.

Table 1: Performance of models on validation data for Task 1.

Model	Dataset	Dataset Size	On test data				
			F1-en	F1-de	F1-fr	F1-ru	F1
Roberta	Training	31,187	0.742	0.631	0.759	0.584	0.671
Mistral	Training	31,187	0.755	0.712	0.721	0.591	0.682
Mistral	Training + Paraphrased FP (Iteration 1)	32,031	0.759	0.701	0.760	0.621	0.695
Mistral	Training + Paraphrased FP (Iterations 1 + 2)	32,894	0.783	0.750	0.783	0.620	0.709
Compared to other teams							
	Mean		0.591	0.557	0.588	0.472	0.539
	Median		0.692	0.669	0.687	0.537	0.627

Abbreviations: en = English, de = German (Deutsch), F1 = F1 Score, FP = False Predictions, ru = Russian.

Table 2: Performance of models on test data for Task 1.

Model	Dataset	Dataset Size	On validation data			On test data		
			Pr	Re	F1	Pr	Re	F1
Roberta	Training	2,521	0.950	0.975	0.962	0.93	0.98	0.95
Mistral	Training	2,521	0.951	0.853	0.899	0.95	0.80	0.87
Mistral	Training + Paraphrased FP (Iteration 1)	2,669	0.959	0.967	0.963	0.96	0.97	0.96
Mistral	Training + Paraphrased FP (Iterations 1 + 2)	2,675	0.962	0.97	0.966	0.93	0.96	0.94
Compared to other teams								
	Mean					0.916	0.961	0.938
	Median					0.922	0.972	0.944

Abbreviations: F1 = F1 Scores, FP = False Predictions, Pr = Precision, Re = Recall.

Table 3: Performance models for Task 6.

scratch. This process was repeated twice for both Task 1 and Task 6. The prompt used for paraphrasing is provided in Appendix C.

As a result, we generated three versions of the training set for each task:

- Original training data
- Original training data + paraphrased false predictions from the first iteration
- Original training data + paraphrased false predictions from both the first and second iterations

Regarding model performance, after the initial fine-tuning step, we applied the iterative error-driven data augmentation strategy to both RoBERTa and Mistral. While the approach showed strong potential for Mistral, applying the same strategy to RoBERTa resulted in no significant improvement on the validation data. As a result, this paper focuses on reporting the outcomes of iterative augmentation applied to Mistral models. Table 1 and Table 2 present the performance results for Task 1 on validation and test data, respectively.

Table 3 shows the performance results for Task 6 on both validation and test data. The best-performing Mistral models on the test data for both Task 1³ and Task 6⁴ have been made publicly available on Hugging Face.

5 Results Analysis

Based on the results of our experiments, we demonstrated that, in contrast to previous studies—such as Wang et al. (2023), Sun et al. (2023), and Khademi et al. (2024)—which showed that LLM-based models often underperform compared to encoder-based models, our iterative data augmentation strategy enabled Mistral to outperform the encoder-based RoBERTa model in both tasks. The effectiveness of our iterative paraphrasing-based augmentation method stems from its ability to exploit hard-to-learn instances. By paraphrasing these examples, we generate semantically equivalent but lexically diverse inputs, which enrich the training data with challenging variations. This encourages

³ Best model_Task 1: <https://huggingface.co/pei-germany/mistral-smm4h2025-task1-ade>

⁴ Best model_Task 6: <https://huggingface.co/pei-germany/mistral-smm4h2025-task6-vaem>

the model to focus on deeper semantic patterns rather than relying on superficial, dataset-specific features. Although this approach led to improved performance overall, paraphrasing very short examples occasionally resulted in outputs that were too similar to the original text, which may have introduced a risk of overfitting (see Appendix D).

Below, we present a detailed breakdown of the results for each task:

5.1 Results for Task 1

In Task 1 (as shown in Table 1 and Table 2), the fine-tuned Mistral model demonstrated slightly better F1 performance across all languages compared to the fine-tuned RoBERTa model on the training data (0.728 vs. 0.726). After applying the first iteration of our iterative augmentation approach, 844 paraphrased false prediction examples (196 false positives and 648 false negatives) were added to the training set, resulting in a slight performance improvement on both the validation and test sets.

In the second iteration, the Mistral model—fine-tuned on the dataset comprising Training + Paraphrased FP (Iteration 1)—was re-evaluated on the training data. This process identified an additional 863 paraphrased false prediction examples (454 false positives and 409 false negatives), which were also incorporated into the training set. Finally, the Mistral model fine-tuned on Training + Paraphrased FP (Iterations 1 + 2) achieved its best performance, with an F1 score of 0.761 on the validation set and 0.709 on the test set.

Moreover, the discovery of 863 remaining false predictions in the second iteration suggests that the model still faced some challenges in learning the training data. Therefore, exploring a third iteration in future work may be worthwhile to assess whether further improvements are achievable or if overfitting would begin to occur.

It is worth mentioning that our model achieved the highest F1 score among all participating systems in Task 1 at the #SMM4H-HeaRD 2025 shared task. As shown in Table 2, our best-performing model—Mistral fine-tuned with paraphrased false predictions from two iterations—achieved a macro-averaged F1 score of 0.709 on the test set. This result is higher than both the median (0.627) and the mean (0.539) F1 scores reported across all participating teams. Furthermore, our model outperformed the median F1 score in all four languages: English (0.783 vs. 0.692), German (0.750 vs. 0.669), French (0.783 vs. 0.687), and Russian (0.620 vs. 0.537). These results highlight the effectiveness of our iterative error-driven data augmentation approach in improving performance across multiple languages.

5.1 Results for Task 6

In Task 6 (as shown in Table 3), after the first iteration of our augmentation approach, 148 paraphrased false prediction examples (comprising 2 false positives and 146 false

negatives) were added to the training data. In the second iteration, an additional 6 paraphrased false predictions (3 false positives and 3 false negatives) were incorporated.

Overall, the Mistral model showed improved performance on the validation set after each iteration, achieving F1 scores of 0.96 and 0.97, respectively. However, on the test set, performance slightly declined—from 0.96 after the first iteration to 0.94 after the second. This drop may be due to overfitting, as only a small number of false predictions (6 examples) were found in the second iteration, indicating that the model may have overfit the training data at this stage.

In summary, the Mistral model fine-tuned on Training + Paraphrased FP (Iteration 1 + Iteration 2) achieved the best result on the validation set, with an F1 score of 0.966, while the model fine-tuned on Training + Paraphrased FP (Iteration 1) performed best on the test set, achieving an F1 score of 0.96.

Due to the submission limit of two models for Task 6 and the unavailability of labels in the test data, we based our selection on validation performance and submitted the following models: (1) the RoBERTa model fine-tuned on the original training data, and (2) the Mistral model fine-tuned on Training + Paraphrased FP (Iteration 1 + Iteration 2). Although our models did not achieve the highest F1 score in Task 6, they performed competitively compared to other participating teams. As shown in Table 3, our best-submitted model achieved an F1 score of 0.94 on the test set—exceeding the mean F1 score of 0.938 and closely approaching the median score of 0.944 reported across all submissions.

6 Conclusion

In this paper, we outlined our contributions to Tasks 1 and 6 of the 2025 SMM4H Workshop, which focused on the binary classification of social media text reporting adverse medical events—multilingual content for Task 1 and English Reddit posts for Task 6. We employed a RoBERTa-based model as a baseline and explored the generative model Mistral. To enhance Mistral’s performance, we applied an iterative error-driven data augmentation strategy, in which false predictions were paraphrased, added back to the training data, and the model was retrained from scratch. Using this approach, we achieved an F1 score of 0.761 on the validation set and 0.709 on the test set for Task 1. For Task 6, the best-performing model achieved an F1 score of 0.966 on the validation set and 0.96 on the test set. We demonstrated that our iterative augmentation strategy enabled Mistral to outperform the encoder-based RoBERTa model in these binary classification challenges. Notably, our system not only achieved the best performance in our own experiments but also obtained the highest F1 score (0.709) on the Task 1 test set among all participating systems. We are currently exploring further enhancements to this method.

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References

- Barbieri, F.; Espinosa Anke, L.; and Camacho-Collados, J. 2022. XLM-T: Multilingual Language Models in Twitter for Sentiment Analysis and Beyond. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, 258–266. Marseille, France: European Language Resources Association.
- Freund, Y.; and Schapire, R. E. 1996. Experiments with a New Boosting Algorithm. In *Proceedings of the Thirteenth International Conference on Machine Learning*, 148–156. San Francisco, CA: Morgan Kaufmann Publishers.
- Khademi, S.; Palmer, C.; Dimaguila, G. L.; Javed, M.; and Buttery, J. 2024. Exploring Large Language Models for Detecting Online Vaccine Reactions. In *Health. Innovation. Community: It Starts With Us*, 30–35. Amsterdam: IOS Press.
- Klein, A. Z.; Banda, J. M.; Guo, Y.; Schmidt, A. L.; Xu, D.; Flores Amaro, I.; Rodriguez-Esteban, R.; Sarker, A.; and Gonzalez-Hernandez, G. 2024. Overview of the 8th Social Media Mining for Health Applications (#SMM4H) Shared Tasks at the AMIA 2023 Annual Symposium. *Journal of the American Medical Informatics Association* 31(4): 991–996.
- Klein, A. Z.; Dasgupta, T.; Flores Amaro, I.; Jana, S.; Khademi, S.; Lopez-Garcia, G.; Onishi, T.; Powell, J.; Raithel, L.; Rajwal, S.; Roller, R.; Sarker, A.; Sinha, M.; Thomas, P.; Tutubalina, E.; Xu, D.; Zweigenbaum, P.; and Gonzalez-Hernandez, G. 2025. Overview of the 10th Social Media Mining for Health (#SMM4H) and Health Real-World Data (HeaRD) Shared Tasks at ICWSM 2025. In *Workshop Proceedings of the 19th International AAAI Conference on Web and Social Media*. Palo Alto, CA: AAAI Press.
- Li, H.; Zhang, Y.; Zhang, Y.; Jiang, S.; and Dong, B. 2024. SRCB at #SMM4H 2024: Making Full Use of LLM-based Data Augmentation in Adverse Drug Event Extraction and Normalization. In *Proceedings of the 9th Social Media Mining for Health Research and Applications (SMM4H 2024) Workshop and Shared Tasks*, 32–37. Bangkok, Thailand: Association for Computational Linguistics.
- Liu, Y.; Ott, M.; Goyal, N.; Du, J.; Joshi, M.; Chen, D.; Levy, O.; Lewis, M.; Zettlemoyer, L.; and Stoyanov, V. 2019.[cs.CL]. Ithaca, NY: Cornell University Library. RoBERTa: A Robustly Optimized BERT Pretraining Approach. *arXiv preprint*. arXiv:1907.11692 [cs.CL]. Ithaca, NY: Cornell University Library.
- Magge, A.; Klein, A.; Miranda-Escalada, A.; Al-Garadi, M. A.; Alimova, I.; Miftahutdinov, Z.; Farre, E.; et al. 2021. Overview of the Sixth Social Media Mining for Health Applications (#SMM4H) Shared Tasks at NAACL 2021. In *Proceedings of the Sixth Social Media Mining for Health (#SMM4H) Workshop and Shared Task*, 21–32. Online: Association for Computational Linguistics.
- Mukans, E.; and Barzdins, G. 2024. RIGA at SMM4H-2024 Task 1: Enhancing ADE Discovery with GPT-4. In *Proceedings of the 9th Social Media Mining for Health Research and Applications (SMM4H 2024) Workshop and Shared Tasks*, 23–27. Bangkok, Thailand: Association for Computational Linguistics.
- Raithel, L.; Yeh, H.-S.; Yada, S.; Grouin, C.; Laverigne, T.; Névélol, A.; Paroubek, P.; Thomas, P.; Nishiyama, T.; Möller, S.; Aramaki, E.; Matsumoto, Y.; Roller, R.; and Zweigenbaum, P. 2024. A Dataset for Pharmacovigilance in German, French, and Japanese: Annotating Adverse Drug Reactions across Languages. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, 395–414. Torino, Italy: ELRA and ICCL.
- Raithel, L.; Thomas, P.; Roller, R.; Sapina, O.; Möller, S.; and Zweigenbaum, P. 2022. Cross-lingual Approaches for the Detection of Adverse Drug Reactions in German from a Patient's Perspective. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, 3637–3649. Marseille, France: European Language Resources Association.
- Sun, X.; Li, X.; Li, J.; Wu, F.; Guo, S.; Zhang, T.; and Wang, G. 2023. Text Classification via Large Language Models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 8990–9005. Singapore: Association for Computational Linguistics.
- Ta, H.-T.; Rahman, A. B. S.; Najjar, L.; and Gelbukh, A. 2024. ThangDLU at #SMM4H 2024: Encoder-decoder Models for Classifying Text Data on Social Disorders in Children and Adolescents. In *Proceedings of the 9th Social Media Mining for Health Research and Applications (SMM4H 2024) Workshop and Shared Tasks*, 1–4. Bangkok, Thailand: Association for Computational Linguistics.
- Tutubalina, E.; Alimova, I.; Miftahutdinov, Z.; Sakhovskiy, A.; Malykh, V.; and Nikolenko, S. 2021. The Russian Drug Reaction Corpus and Neural Models for Drug Reactions and Effectiveness Detection in User Reviews. *Bioinformatics* 37(2): 243–249. <https://doi.org/10.1093/bioinformatics/btaa675>.
- Wang, S.; Sun, X.; Li, X.; Ouyang, R.; Wu, F.; Zhang, T.; Li, J.; and Wang, G. 2023. GPT-NER: Named Entity Recognition via Large Language Models. *arXiv preprint*. arXiv:2304.10428 [cs.CL]. Ithaca, NY: Cornell University Library.
- Xu, D.; Lopez-Garcia, G.; Raithel, L.; Roller, R.; Thomas, P.; Aramaki, E.; Wakamiya, S.; et al. 2024. Overview of the 9th Social Media Mining for Health Applications (#SMM4H) Shared Tasks at ACL 2024–Large Language Models and Generalizability for Social Media NLP. In *Proceedings of the 9th Social Media Mining for Health Research and Applications (SMM4H 2024) Workshop and Shared Tasks*, 183–195. Bangkok, Thailand: Association for Computational Linguistics.

A Classification Prompt for Task 1

For classification in Task 1, we used the following prompt:

Task:

Classify whether the given text mentions an "Adverse Drug Event".

Definition:

An "Adverse Drug Event" is an observation made by "patients or clinicians" about "physical or mental abnormalities" suspected to be caused by "medications or vaccines".

Guidelines:

- The text is a tweet/post where a doctor or patient discusses a medical condition.
- Patients may indirectly imply they are using a medication that caused an ADE.
- If an ADE is explicitly or implicitly mentioned, classify it as 1 (Positive).
- If there is no mention of an ADE, classify it as **0 (Negative).

Output Format:

Return only:

1 → ADE mentioned

0 → No ADE mentioned

Input:

B Classification Prompt for Task 6

For classification in Task 6, we used the following prompt:

Task

Determine whether the given text mentions an Adverse Drug Event (ADE) associated with a herpes zoster (shingles) vaccine.

Definitions

- Adverse Drug Event (ADE): A physical or mental abnormality observed by a patient or clinician and suspected to be caused by a medication or vaccine.
- Zoster vaccine: A vaccine intended to prevent herpes zoster (shingles), caused by the reactivation of the varicella-zoster virus.

Classification Guidelines

- The input is typically a social media post (e.g., tweet) authored by a patient or healthcare professional.
- Indirect mentions of vaccine use or ADEs (e.g., implied through symptoms or timing) should be considered.
- Classify as 1 (Positive) if:
 - An ADE is explicitly mentioned or strongly implied
 - There is a clear association with a zoster vaccine
- Classify as 0 (Negative) if:
 - There is no mention of a zoster vaccine, or
 - There is no mention (explicit or implied) of an ADE

Output Format

Return only one of the following:

- 1 → ADE related to a zoster vaccine is mentioned

- 0 → No ADE or no link to a zoster vaccine

Input:

C Paraphrasing Prompt

For paraphrasing, we used the following prompt:

Task:

Paraphrase the input text, which discusses a possible Adverse Drug Event (ADE).

Definition:

An Adverse Drug Event (ADE) refers to physical or mental abnormalities reported by patients or clinicians that are suspected to be caused by medications or vaccines.

Guidelines:

- The text is a tweet or post where a doctor or patient describes a medical condition.
- It may contain informal language, typos, or grammatical errors.
- Do not change the original language of the input text.

Input:

D Original and Paraphrased Instances Used in Error-Driven Augmentation

Below are examples of false predictions and their corresponding paraphrased versions generated through our augmentation strategy:

Task 1 Example:

#Original (True label = 1, Predicted label = 0):

"@USER my doctor gave me paxil, which i think is similar but it's made me feel like crap."

Paraphrased Version:

"I started taking Paxil on my doc's advice, but it's been wreaking havoc on me - not feeling great at all."

Task 6 Example:

#Original (True label = 0, Predicted label = 1):

"Going to get my 2nd shingles shot in 15 minutes, witness me!: Oof! I have had shingles twice within two years."

#Paraphrased Version:

"About to get my second shingles vaccine in 15 minutes - wish me luck! :Ouch: I've had shingles twice in the last two years."